Income and Inflation Analysis using Linear Regression and Predictive Modeling (1984–2024)

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DAT 301 Project 1

Contents

Introduction **1**

Formal Data Descriptions and Sources **2**

Research Questions **3**

Exploratory Data Analysis **4**

Indexed comparison (fixed base year) 5

Regression Analysis: Income ~ CPI **6**

Fitting the model 6

Scatterplot with regression line 7

Residual diagnostics 8

Predictive Modeling Beginning at the 2008 Financial Crisis **10**

Model Training and Prediction Code 10

Actual vs. Predicted Plot 10

Interpretation 11

Conclusions **11**

Appendix: Full Data Table (Optional) **12**

Introduction

In this project, I analyze how median household income in the United States has changed relative to inflation, measured by the Consumer Price Index (CPI), over the period 1984–2024.

The problem I'm trying to solve:

- With inflation dominating recent economic and political conversations, the goal of this project was to compare how people perceive inflation to how it actually behaves according to official data from the U.S. Bureau of Labor Statistics. To explore this gap, I collected perception data from individuals in my surroundings and evaluated it against the market-basket price levels measured by the Consumer Price Index (CPI). This comparison allowed me to assess whether personal impressions of inflation align with the actual trends reflected in our economic statistics.

This analysis is based on publicly available data, published by the US Bureau of Labor Statistics and the US Census Bureau. My analysis demonstrates data manipulation (through cleaning of the data sets to index average salary through the years to the average of 1984), visualization (graphs), and statistical modeling in R (utilizing linear regression).

Formal Data Descriptions and Sources

- **Median Household Income (MHI)**
  - Source: US Census Bureau
  - Variable: annual U.S. median household income in dollars
  - File used: `mhi.csv`
- **Consumer Price Index (CPI-U)**
  - Source: U.S. Bureau of Labor Statistics (BLS)
  - Variable: CPI for All Urban Consumers (CPI-U), monthly
  - File used: `cpi-u.csv`
  - I aggregate the monthly series to a yearly average.

Both CSV files are in the same working directory as my R Markdown file

```r
# Read median household income csv
mhi <- read.csv("mhi.csv", stringsAsFactors = FALSE)
names(mhi) <- c("observation_date", "Income")
mhi$Year <- as.numeric(substr(mhi$observation_date, 1, 4))

# Read CPI data csv
cpi <- read.csv(
  "cpi-u.csv",
  header = TRUE,
  sep = "",
  quote = "",
  stringsAsFactors = FALSE
)

# keep only monthly observations (period starting with 'M')
# helps keep the yearly observations yearly
cpi <- cpi %>% dplyr::filter(grepl("^M", period))
cpi$year <- as.numeric(cpi$year)

# change CPI to yearly averages
```

```r
cpi_yearly <- cpi %>%
  dplyr::group_by(year) %>%
  dplyr::summarise(CPI = mean(value, na.rm = TRUE)) %>%
  dplyr::rename(Year = year)

# clean mhi and join with CPI data
mhi_clean <- mhi %>% dplyr::select(Year, Income)

data_all <- mhi_clean %>%
  dplyr::inner_join(cpi_yearly, by = "Year") %>%
  dplyr::arrange(Year)

head(data_all)
```

```
##   Year Income      CPI
## 1 1984  60420 187.2958
## 2 1985  61570 193.9955
## 3 1986  63850 197.4099
## 4 1987  64650 183.9534
## 5 1988  65130 191.4889
## 6 1989  66240 200.8595
```

Variables + Data Cleaning:

The combined dataset `data_all` contains:

- `Year`: calendar year
- `Income`: median household income in USD
- `CPI`: annual average CPI for all urban consumers

I restricted the analysis to the years I have both income and CPI (1984 to 2024)

```r
range(data_all$Year)
```

```
## [1] 1984 2024
```

```r
nrow(data_all)
```

```
## [1] 41
```

Research Questions

I focused on two main questions to answer my main problem:

1. **Is there a significant linear relationship between CPI and median household income?**

I modeled `Income ~ CPI` using linear regression and assessed whether CPI is a useful predictor of actual income.

2. **Has median household income kept pace with inflation when both series are indexed to a base year of 1984?**

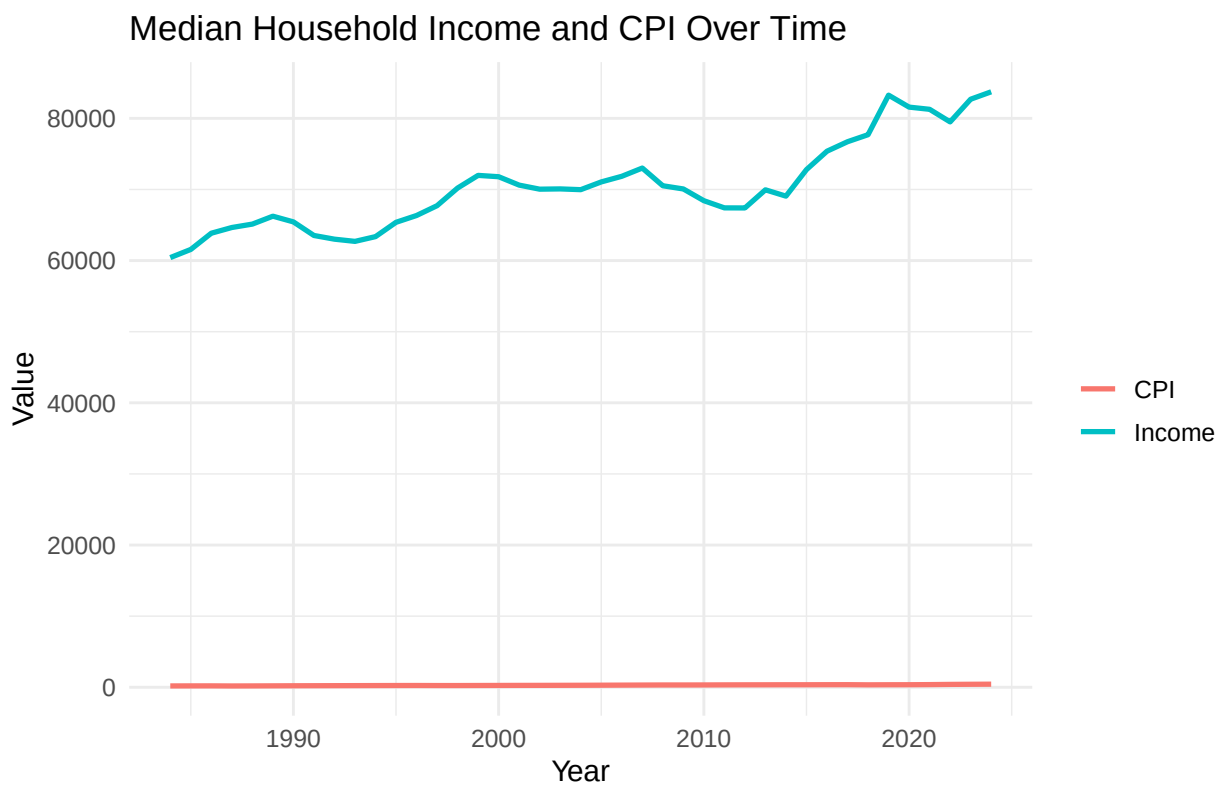
I constructed income and CPI indexes with a chosen base year (index = 100) and compared their trajectories over time. I specifically chose 1984 because the indexes for CPI is based off of the market basket from 1982 to 1984. These questions help to understand whether rising incomes are simply keeping up with inflation, or whether purchasing power has meaningfully changed throughout the years.

Exploratory Data Analysis

First, I visualized income and CPI over time to see their overall trends.

```
data_long <- data_all %>%
  tidyr::pivot_longer(
    cols = c("Income", "CPI"),
    names_to = "Variable",
    values_to = "Value"
  )

ggplot(data_long, aes(x = Year, y = Value, color = Variable)) +
  geom_line(linewidth = 1) +
  theme_minimal(base_size = 12) +
  labs(
    title = "Median Household Income and CPI Over Time",
    x = "Year",
    y = "Value",
    color = ""
  )
```



Both series trend upward over time, but they are on different scales (dollars vs. index units), so this doesn't tell us much

Indexed comparison (fixed base year)

To make income and CPI more comparable, I constructed indexes relative to the aforementioned base year of 1984. I set the base year to 1984 so that the index directly coincides with the cpi market basket as well.

This way a change of 2x for CPI is definately comparable to a change of 1.5x for household income.

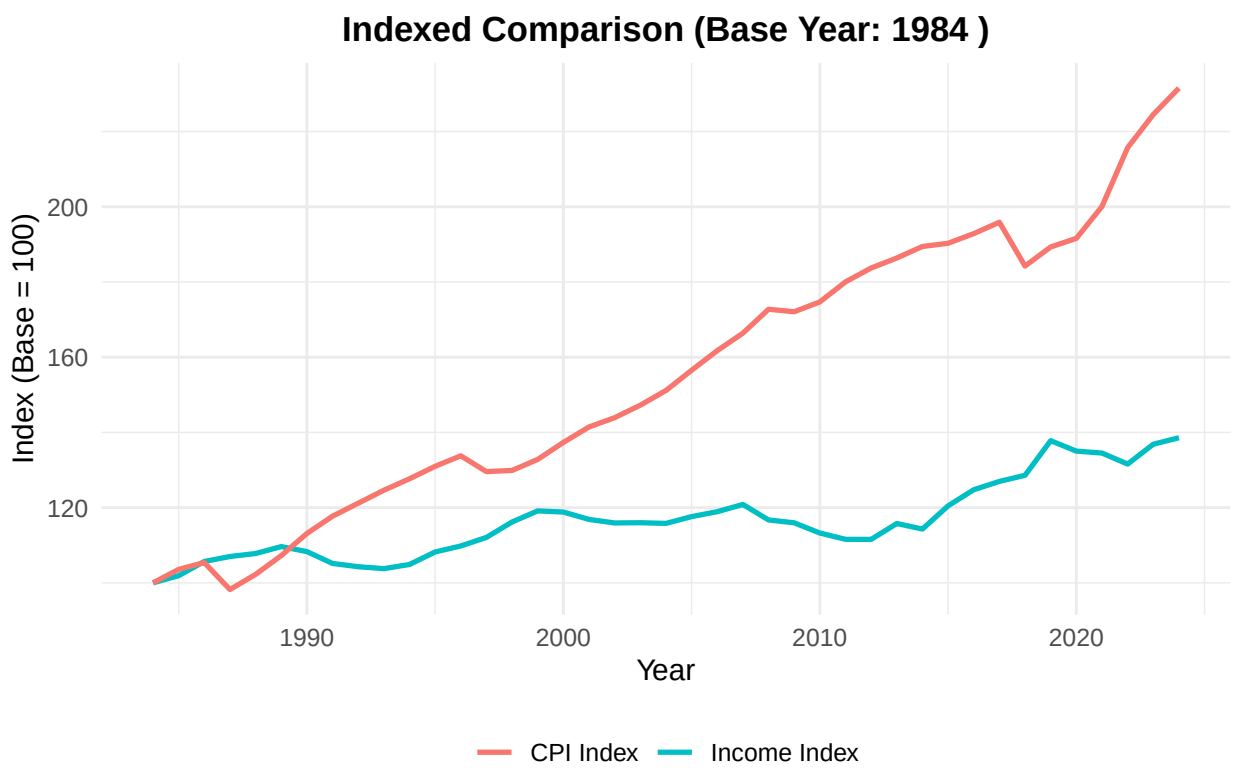
```
base_year <- min(data_all$Year)

base_income <- data_all$Income[data_all$Year == base_year]
base_cpi    <- data_all$CPI[data_all$Year == base_year]

data_indexed <- data_all %>%
  dplyr::mutate(
    IncomeIndex = 100 * Income / base_income,
    CPIIndex    = 100 * CPI / base_cpi
  )

ggplot(data_indexed, aes(x = Year)) +
  geom_line(aes(y = IncomeIndex, color = "Income Index"), linewidth = 1) +
  geom_line(aes(y = CPIIndex, color = "CPI Index"), linewidth = 1) +
  theme_minimal(base_size = 12) +
  theme(
    plot.title = element_text(hjust = 0.5, face = "bold"),
    legend.title = element_blank(),
```

```
    legend.position = "bottom"
  ) +
  labs(
    title = paste("Indexed Comparison (Base Year:", base_year, ")"),
    y = "Index (Base = 100)"
  )
```



In this indexed plot, if the Income Index consistently stays above the CPI Index, it suggests that income has grown faster than inflation, and vice versa if CPI is above income consistently.

Regression Analysis: Income ~ CPI

Fitting the model

I now fit a simple linear regression model with Income as the response and CPI as the predictor.

```
model_fit <- lm(Income ~ CPI, data = data_all)
summary(model_fit)
```

```
##
## Call:
## lm(formula = Income ~ CPI, data = data_all)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
```

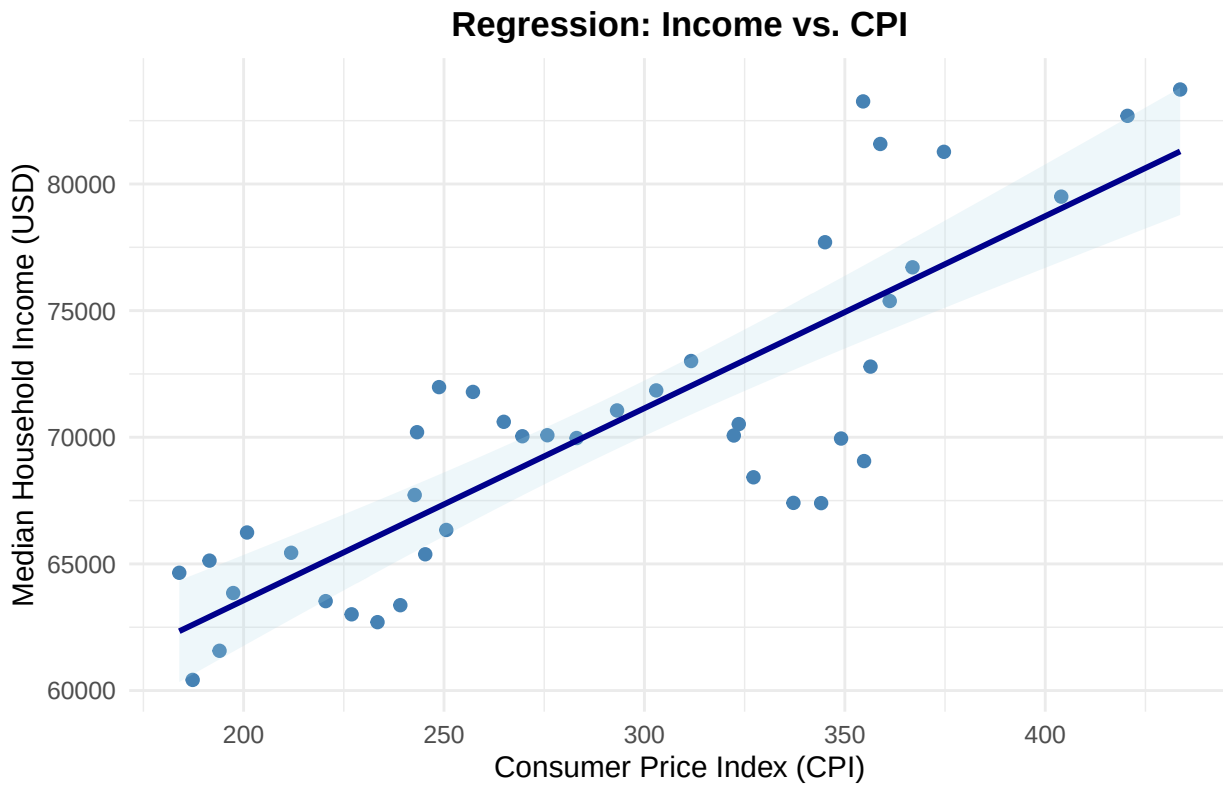
```
## -7090.8 -2411.8   480.4  2307.8  7974.7
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) 48386.78    2333.83  20.733   <2e-16 ***
## CPI            75.86       7.82   9.701    6e-12 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3428 on 39 degrees of freedom
## Multiple R-squared:  0.707,  Adjusted R-squared:  0.6995
## F-statistic: 94.11 on 1 and 39 DF,  p-value: 5.999e-12
```

The summary output includes the estimated intercept and slope, their standard errors and p-values, and overall model fit metrics such as R-squared.

A statistically significant positive slope for CPI suggests that higher CPI levels are associated with higher median household income.

Scatterplot with regression line

```r
ggplot(data_all, aes(x = CPI, y = Income)) +
  geom_point(color = "steelblue", size = 2) +
  geom_smooth(
    method = "lm",
    se = TRUE,
    color = "darkblue",
    fill = "lightblue",
    alpha = 0.2
  ) +
  theme_minimal(base_size = 12) +
  theme(
    plot.title = element_text(hjust = 0.5, face = "bold")
  ) +
  labs(
    title = "Regression: Income vs. CPI",
    x = "Consumer Price Index (CPI)",
    y = "Median Household Income (USD)"
  )
```



The scatterplot shows a clear upward trend, indicating a strong positive relationship between CPI and income. The regression line approximates the average relationship across all years.

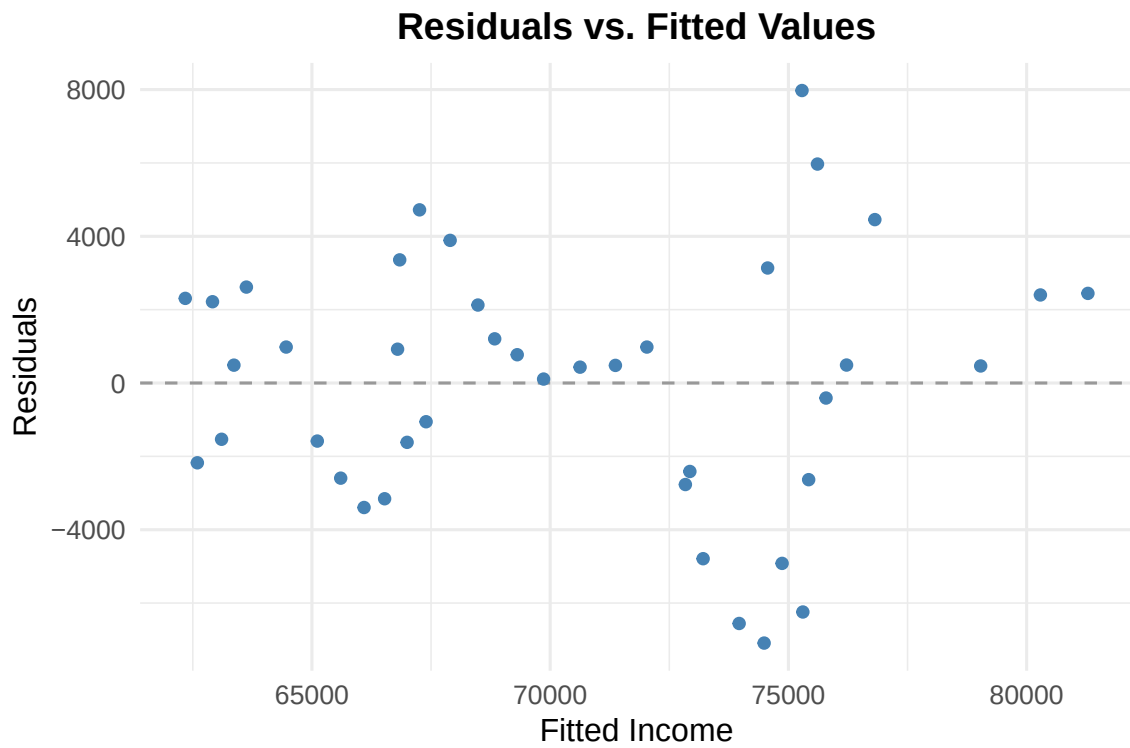
It is of note that the scatterplot has a near stationary trend line post-COVID19. It is also of note that the scattplot actually trended downward immediately following the Great 2008 Recession, up until around 2015.

Residual diagnostics

To check how well the linear model fits the data, I examined residuals.

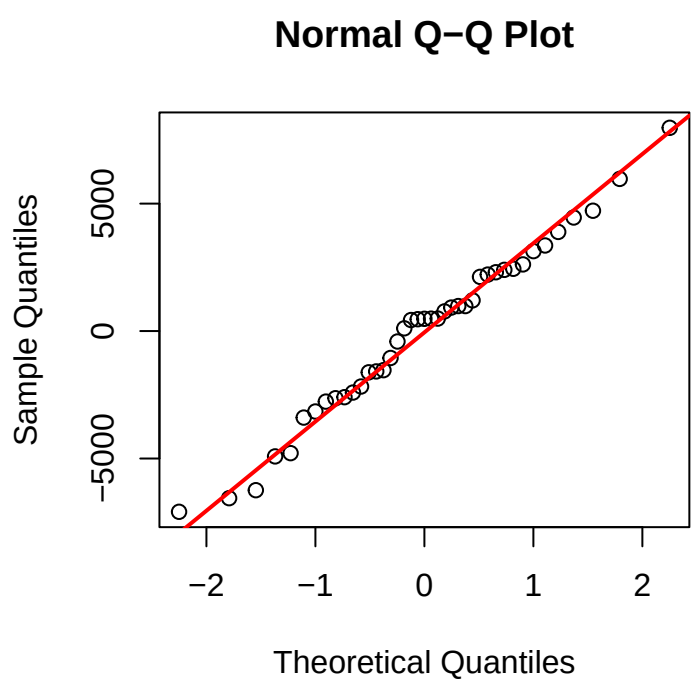
```
df_res <- data.frame(
  Fitted = fitted(model_fit),
  Residuals = resid(model_fit)
)

ggplot(df_res, aes(x = Fitted, y = Residuals)) +
  geom_hline(yintercept = 0, linetype = "dashed", color = "grey60") +
  geom_point(color = "steelblue") +
  theme_minimal(base_size = 12) +
  theme(
    plot.title = element_text(hjust = 0.5, face = "bold")
  ) +
  labs(
    title = "Residuals vs. Fitted Values",
    x = "Fitted Income",
    y = "Residuals"
  )
```



I also look at a normal Q–Q plot of residuals:

```
qqnorm(resid(model_fit))
qqline(resid(model_fit), col = "red", lwd = 2)
```



Overall, the residual diagnostics help assess whether the linear model is appropriate and whether there are particular time periods or economic events that deviate strongly from the fitted trend.

The residual diagnostics show that the linear model captures the overall relationship between CPI and income reasonably well because most residuals fall close to zero.

However, the residuals vs. fitted plot reveals a slight curvature, indicating that the relationship is not perfectly linear across all decades. Large negative residuals appear during the 2008–2013 financial crisis, showing that income dropped more sharply than the inflation-based model would predict.

The Q–Q plot confirms that while most residuals follow a normal distribution, the tails deviate due to outliers associated with major economic shocks such as the financial crisis and the COVID-19 pandemic.

Overall, the diagnostics suggest that the model works well for long-term trends but struggles to account for extreme disruptions in the economy.

Predictive Modeling Beginning at the 2008 Financial Crisis

The 2008 financial crisis introduced a major structural break in the relationship between income and inflation. To evaluate how the crisis disrupted long-term trends, I trained the linear regression model using only the years prior to 2008 and then used this model to generate predicted income values for 2008–2024.

These predictions represent the income levels that would have been expected if the historical inflation–income relationship observed before the crisis had continued without interruption.

Model Training and Prediction Code

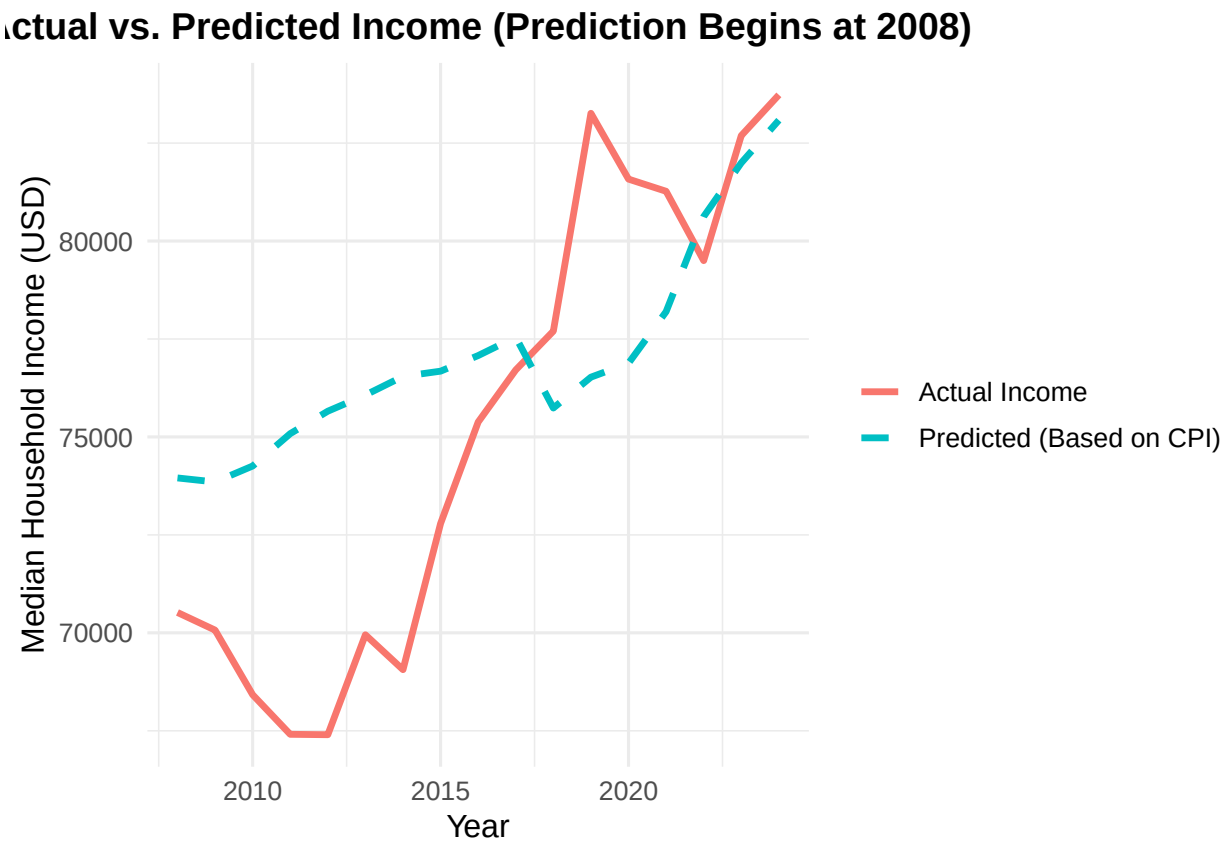
```r
# Split data into pre-crisis (training) and post-crisis (testing)
train <- data_all %>% filter(Year < 2008)
test  <- data_all %>% filter(Year >= 2008)

# Fit model using only pre-2008 data
model_pre2008 <- lm(Income ~ CPI, data = train)

# Predict income for 2008-2024 based on CPI
test$PredictedIncome <- predict(model_pre2008, newdata = test)
```

Actual vs. Predicted Plot

```r
ggplot(test, aes(x = Year)) +
  geom_line(aes(y = Income, color = "Actual Income"), size = 1.2) +
  geom_line(aes(y = PredictedIncome, color = "Predicted (Based on CPI)"),
            size = 1.2, linetype = "dashed") +
  theme_minimal(base_size = 12) +
  theme(
    legend.title = element_blank(),
    plot.title = element_text(hjust = 0.5, face = "bold")
  ) +
  labs(
    title = "Actual vs. Predicted Income (Prediction Begins at 2008)",
    y = "Median Household Income (USD)"
  )
```



Interpretation

The regression model trained on pre-2008 data predicts a relatively smooth increase in income from 2008 onward, consistent with continued long-term inflation. However, actual income levels fall noticeably below the predicted curve from 2008 through approximately 2013, reflecting the prolonged effects of the financial crisis on household earnings.

Between 2015 and 2019, actual income draws closer to the predicted trajectory, indicating partial recovery and a return to the pre-crisis inflation-linked trend. This convergence is disrupted again after 2020, when both the COVID-19 pandemic and rapid inflation in 2021–2022 create new divergences between actual and predicted values.

Overall, this predictive analysis illustrates how simple inflation-based models perform in stable periods but break down sharply during major economic shocks. The comparison highlights the lasting impact of the 2008 recession and the volatility of the post-COVID recovery, demonstrating the limits of relying solely on CPI to predict income trends.

Conclusions

Based on the analysis above:

1. **Strong positive relationship between CPI and income.**
2. **Income and inflation do not move together and things have gotten substantially more expensive for the average American**

3. **The economy is volatile and unpredictable, usually being affected by things outside of most people's control**

Overall, this project shows that median household income has generally increased alongside inflation over the past several decades, but the relationship is not perfectly one-to-one, and key economic events can create noticeable gaps between income growth and price growth.

Appendix: Full Data Table (Optional)

```
##    Year Income      CPI
## 1  1984  60420 187.2958
## 2  1985  61570 193.9955
## 3  1986  63850 197.4099
## 4  1987  64650 183.9534
## 5  1988  65130 191.4889
## 6  1989  66240 200.8595
## 7  1990  65440 211.8659
## 8  1991  63530 220.4783
## 9  1992  63010 226.9502
## 10 1993  62700 233.4121
## 11 1994  63370 239.0998
## 12 1995  65380 245.3207
## 13 1996  66340 250.5676
## 14 1997  67720 242.6857
## 15 1998  70200 243.2719
## 16 1999  71980 248.7620
## 17 2000  71790 257.2238
## 18 2001  70610 264.9116
## 19 2002  70040 269.5477
## 20 2003  70080 275.7836
## 21 2004  69970 283.0820
## 22 2005  71060 293.1696
## 23 2006  71850 302.9484
## 24 2007  73010 311.6491
## 25 2008  70520 323.5411
## 26 2009  70070 322.3034
## 27 2010  68420 327.2095
## 28 2011  67410 337.1776
## 29 2012  67400 344.0904
## 30 2013  69950 349.0618
## 31 2014  69060 354.8015
## 32 2015  72790 356.4092
## 33 2016  75380 361.2103
## 34 2017  76710 366.8892
## 35 2018  77700 345.0583
## 36 2019  83260 354.5631
## 37 2020  81580 358.8480
## 38 2021  81270 374.7225
## 39 2022  79500 403.9961
## 40 2023  82690 420.5196
## 41 2024  83730 433.6636
```